Index Equals Value

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The problem

- Sorted distinct ints; smallest index i with a[i] == i, or -1.
- Binary search.

Intuition

- Since values are distinct sorted integers, value minus index never decreases, so binary search works.

Brute force vs optimal

- Linear scan → O(n)
- Binary search on a[i] – i (non-decreasing) → O(log n)
- Keep searching left for the first

Approach

- 1. Binary search
- 2. If the value is below the index, go right
- 3. Otherwise record a match and go left for the smallest

Worked example




```
a[3] = 3 = 3 → record, look left
```

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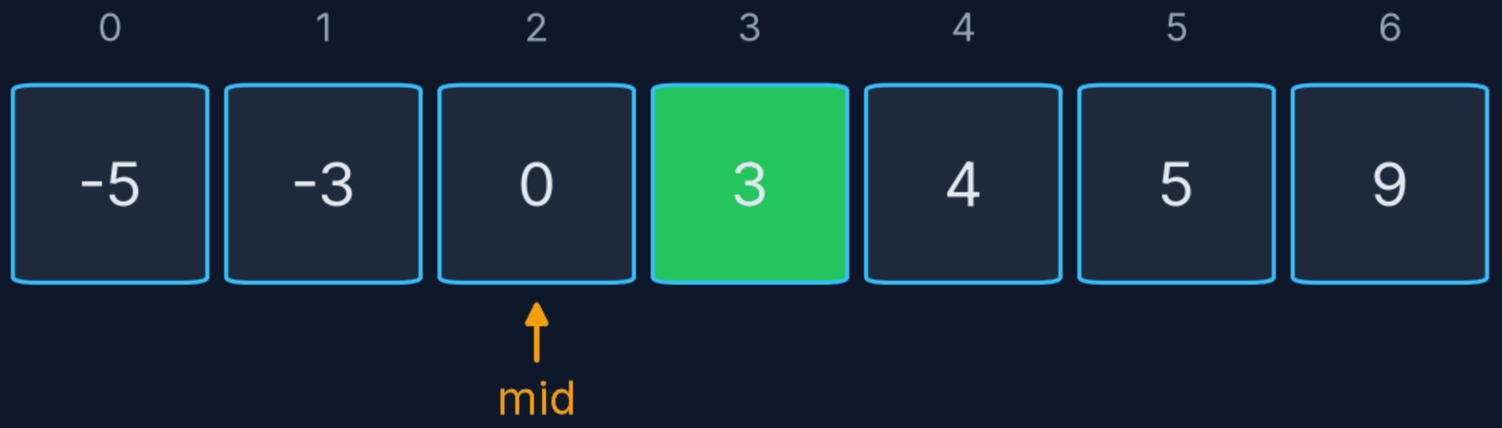
Worked example




```
a[1] = -3 < 1 → go right
```

a[1] = -3 < 1 → go right

Worked example




```
a[2] = 0 < 2 → go right
```

a[2] = 0 < 2 → go right

Java solution

```java
public class IndexEqualsValue {
    public static int indexEqualsValue(int[] a) {
        int lo = 0, hi = a.length - 1, ans = -1;
        while (lo <= hi) {
            int m = (lo + hi) >>> 1;
            if (a[m] < m) lo = m + 1;
            else { if (a[m] == m) ans = m; hi = m - 1; }
        }
        return ans;
    }
}
```

Complexity

- Time: O(log n)
- Space: O(1)

Edge cases & pitfalls

- Return the first such index
- None → −1
- Distinct integers required

Interview tips & follow-ups

- Explain why a[i] – i is monotone

Recap

- Since values are distinct sorted integers, value minus index never decreases, so binary search works.
- Complexity: O(log n)